

ESCUELA POLITÉCNICA SUPERIOR

INTRODUCTION TO FLUID MECHANICS. PROBLEM 2 (10 POINTS).

14-11-2016
Time: 60 minutes

NAME:.....

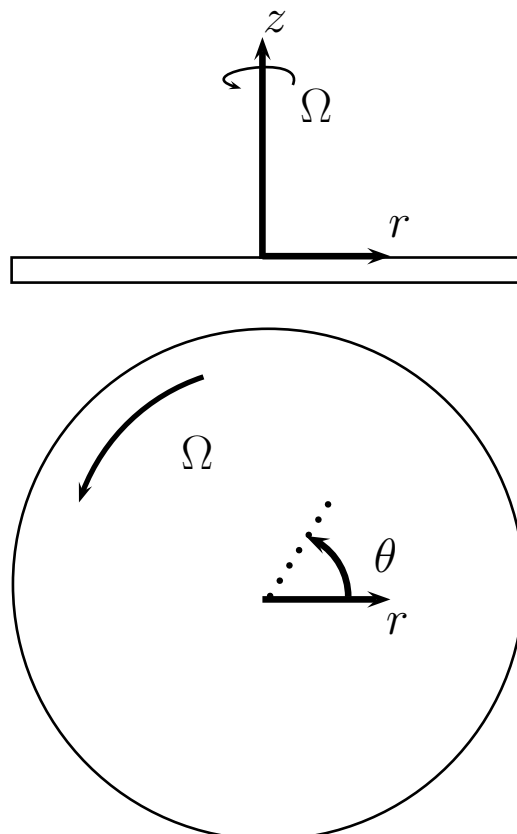
GROUP:.....

We will consider an infinitely large porous disk rotating with angular velocity Ω that sucks the fluid with a known velocity $v_z = -2(\nu\Omega)^{1/2}\lambda_1$. The velocity induced in the fluid of density ρ and kinematic viscosity $\nu = \mu/\rho$ is three-dimensional, with components (v_r, v_θ, v_z) in the cylindrical coordinates system shown in the sketch.

We want to show that the solution of the problem has the form $v_z = -2(\nu\Omega)^{1/2}f(\eta)$, $v_\theta = r\Omega g(\eta)$, where $\eta = (\Omega/\nu)^{1/2}z$ is a similarity variable and $f(\eta)$ and $g(\eta)$ are two functions that must be obtained in the course of the solution, such that the solution far from the disk, as $\eta \rightarrow \infty$, is given by $v_r \rightarrow 0$, $v_\theta \rightarrow 0$.

Taking into account the azimuthal symmetry of the problem,

1. Use the continuity equation to obtain the radial component v_r in terms of the function $f(\eta)$.
2. Use the azimuthal momentum equation to find the ordinary differential equation to be satisfied by the function $g(\eta)$.
3. Use the physical boundary conditions for $\bar{v}$ to find the boundary conditions that $f(\eta)$ and $g(\eta)$ must satisfy.
4. Use the vertical momentum equation to obtain the pressure distribution in terms of an integration function $C(r)$ that will appear in the course of the integration.
5. Use the radial momentum equation to determine $C(r)$ and the ordinary differential equation to be satisfied by $f(\eta)$, with its corresponding boundary conditions. Use the variable separation method to show that $C(r) = A$, where A is a constant. Obtain A in terms of the pressure far from the plate p_∞ and the pressure distribution $p(\eta)$.



$$v_z = -2(\nu r)^{1/2} f(\eta) \quad \text{where } \eta = \left(\frac{r}{\nu}\right)^{1/2} z$$

$$v_\theta = r \Omega g(\eta)$$

① CONTINUITY: $\frac{1}{r} \frac{d}{d\eta}(r v_r) + \frac{dv_z}{dz} = 0 \Rightarrow \boxed{v_r = r \Omega f'(\eta)}$

② θ -MOMENTUM: $\nu r \frac{dv_\theta}{dr} + \frac{\nu v_\theta}{r} + \nu r \frac{dv_\theta}{dz} = \mu \frac{d^2 v_\theta}{dz^2}$

$$5 r \Omega^2 g + 5 r \Omega^2 f' r \Omega g - 2(\nu r)^{1/2} f \cdot \left(\frac{r}{\nu}\right)^{1/2} g' = \mu r \Omega g'' \frac{r}{\nu}$$

$$\boxed{g'' + 2(f g' - f' g) = 0}$$

③ B.C: $z=0: v_r = 0 = r \Omega f'(0)$
 $v_\theta = \Omega r = r \Omega g(0)$
 $v_z = -2(\nu r)^{1/2} \lambda_1 = -2(\nu r)^{1/2} f(0)$

$$\left. \begin{array}{l} f(0) - \lambda_1 = f'(0) = 0 \\ f'(\infty) = 0 \\ g(0) = 1 \\ g(\infty) = 0 \end{array} \right\}$$

$z \rightarrow \infty: v_r = 0 \Rightarrow f'(\infty) = 0$
 $v_\theta = 0 \Rightarrow g(\infty) = 0$

④ Z-MOMENTUM: $\nu r \frac{dv_z}{dz} = -\frac{dp}{dz} + \mu \frac{d^2 v_z}{dz^2} \rightarrow \frac{d}{dz} \left(p + \frac{1}{2} \nu r^2 + \mu \frac{dv_z}{dz} \right) = 0$

$$p + \frac{1}{2} \nu r^2 = \mu \frac{dv_z}{dz} = C(r) \rightarrow \boxed{p = C(r) - 2\mu \Omega [f^2 + f']}$$

⑤ r-MOMENTUM: $\nu r \frac{dv_r}{dr} - \frac{\nu v_\theta^2}{r} + \nu r \frac{dv_r}{dz} = -\frac{\partial p}{\partial r} + \mu \frac{d^2 v_r}{dz^2}$

$$5 r \Omega^2 f^{1/2} - 5 r \Omega^2 g^2 - 2(\nu r)^{1/2} f \cdot r \Omega \left(\frac{r}{\nu}\right)^{1/2} f'' = -C' + \mu r \Omega f''' \frac{r}{\nu}$$

$$\underbrace{-f'' + f^{1/2} - g^2 - 2ff''}_{\text{FUNCTION OF } \eta} = \underbrace{-\frac{C'(r)}{5 r \Omega^2}}_{\text{FUNCTION OF } r} \stackrel{\uparrow \text{NECESSARILY}}{=} \text{CONST} = -\lambda$$

as $\eta \rightarrow \infty: f' \rightarrow 0 \Rightarrow f'' \rightarrow 0 \Rightarrow f''' \rightarrow 0 \Rightarrow \lambda = 0 \Rightarrow C'(r) = 0 \Rightarrow \boxed{C(r) = A}$

$g \rightarrow 0$

as $\eta \rightarrow \infty: p = p_\infty = A - 2\mu \Omega [f(\infty)^2 - f'(\infty)] \Rightarrow A = p_\infty + 2\mu \Omega f(\infty)^2$

$$\boxed{p = p_\infty - 2\mu \Omega [f^2(\eta) - f'(\infty) + f'(\eta)]}$$